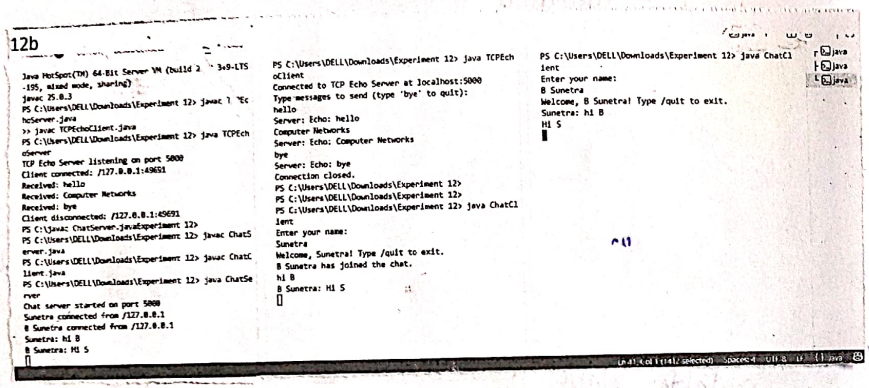


ROUTING AT NETWORK LAYER

AIM :-

To simulate and configure static routing protocol configuration using Cisco Packet Tracer and to verify communication between networks.

ALGORITHM :

1. In Cisco Packet Tracer, place 2 routers, 2 switches and PC.
2. Connect PC to switch & switches to router using Ethernet Cables
3. Assign IP addresses to all PC and router interfaces.
4. Configure default gateway for each PC.
5. Configure static routes in both routers using "ip route command
6. Activate all router interfaces using "no shutdown".
7. Check the routing table using "show ip route".
8. Test connectivity between PC in different networks using ping.
9. Use Simulation Mode to Observe Path.

PROCEDURE :

1. Create the topology
2. Configure PC — IP address : 192.168.1.2
   Subnet Mask : 255.255.255.0
   Default Gateway : 192.168.1.1.
3. Configure Router-1

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
```

interface gigabitEthernet 0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit.

6. On the routers : show ip route,
from Pc1 : ping 192.168.2.2 .

Pc1 → Switch 1 → Router 1 → Router 2 → Switch 2 → PC 2 .

RESULT :

Thus, static routing was successfully configured and simulated using Cisco Packet Tracer